HOMEWORK 2 - INTRODUCTION TO ELLIPTIC CURVES - AUGUST-NOVEMBER 2026 - NOT TO BE SUBMITTED

AEC below refers to the book "The Arithmetic of Elliptic Curves" by Joseph Silverman.

(1) AEC, 2.1.

(2) AEC, 2.2.

(3) AEC, 2.4.

(4) AEC, 2.7.

(5) AEC, 2.8.

(6) AEC, 2.10.

(7) AEC, 2.14.

(8) AEC, 2.15.

(9) (Riemann-Hurwitz formula) Let $\phi : C_1 \longrightarrow C_2$ be a nonconstant separable morphism of smooth curves. For $P \in C_1$, put $Q = \phi(P)$. If t is a uniformizer at Q, define

$$d_\phi(P) = \operatorname{ord}_P\big(d(\phi^* t)\big).$$

The integer $d_\phi(P)$ is called the *different exponent of ϕ at P*.

(a) Prove that $d_\phi(P)$ is independent of the choice of uniformizer t at Q.

(b) Deduce that

$$d_\phi(P) \geq e_\phi(P) - 1.$$

Prove that equality holds whenever $\operatorname{char}(K)$ does not divide $e_\phi(P)$.

(c) Prove that

$$\operatorname{ord}_P(\phi^*\omega) = e_\phi(P) \operatorname{ord}_Q(\omega) + d_\phi(P).$$

(d) Deduce that $d_\phi(P) = 0$ for all but finitely many points $P \in C_1$. Hence the divisor $\operatorname{Diff}(\phi) = \sum_{P \in C_1} d_\phi(P)P$ is well-defined and effective. Prove the divisor identity

$$\operatorname{div}(\phi^*\omega) = \phi^* \operatorname{div}(\omega) + \operatorname{Diff}(\phi).$$

(e) Let K_{C_1} and K_{C_2} be canonical divisors. Deduce that

$$K_{C_1} \sim \phi^* K_{C_2} + \operatorname{Diff}(\phi).$$

(f) Take degrees in the preceding divisor relation to prove the Riemann-Hurwitz formula

$$2g(C_1) - 2 = (\deg \phi)\big(2g(C_2) - 2\big) + \deg \operatorname{Diff}(\phi).$$

(g) Suppose that ϕ is tamely ramified; that is, the characteristic of K does not divide $e_\phi(P)$ for any P. Prove that

$$\operatorname{Diff}(\phi) = \sum_{P \in C_1} \big(e_\phi(P) - 1\big)P,$$

and hence

$$2g(C_1) - 2 = (\deg \phi)\big(2g(C_2) - 2\big) + \sum_{P \in C_1} \big(e_\phi(P) - 1\big).$$

(h) Let $\psi : C_2 \to C_3$ be another nonconstant separable morphism of smooth curves. Prove the local transitivity formula

$$d_{\psi\circ\phi}(P) = d_\phi(P) + e_\phi(P) d_\psi(\phi(P)).$$

Deduce that

$$\operatorname{Diff}(\psi\circ\phi) = \operatorname{Diff}(\phi) + \phi^* \operatorname{Diff}(\psi).$$

(10) This problem is about tame and wild ramification on $\mathbb{P}^1$.

(a) Let $n \geq 2$, assume that $\operatorname{char}(K) \nmid n$, and consider

$$\phi_n : \mathbb{P}^1 \longrightarrow \mathbb{P}^1, \qquad [X : Z] \longmapsto [X^n : Z^n].$$

Determine every ramification index and every different exponent. Compute $\operatorname{Diff}(\phi_n)$ and verify the Riemann–Hurwitz formula.

(b) Now assume that $\operatorname{char}(K) = p > 0$ and consider the Artin–Schreier morphism

$$\psi : \mathbb{P}^1 \longrightarrow \mathbb{P}^1, \qquad t \longmapsto t^p - t.$$

Prove that ψ is separable of degree p. Show that it is unramified at every finite point and that

$$e_\psi(\infty) = p.$$

If $s = 1/t$ is a uniformizer at the source point ∞ and u is the standard uniformizer at the target point ∞, show that

$$\psi^* u = \frac{s^p}{1 - s^{p-1}}.$$

Compute $d_\psi(\infty)$, determine $\operatorname{Diff}(\psi)$, and verify Riemann–Hurwitz. Explain explicitly why

$$d_\psi(\infty) > e_\psi(\infty) - 1.$$

(c) In characteristic p, consider the Frobenius morphism

$$F : \mathbb{P}^1 \longrightarrow \mathbb{P}^1, \qquad t \longmapsto t^p.$$

Explain why the definition of the different divisor used in the preceding problem, and hence the separable form of Riemann–Hurwitz proved there, cannot be applied directly to F.

(11) (Odd degree hyperelliptic curve) Assume that $\operatorname{char}(K) \neq 2$. Let $f(x) \in K[x]$ be squarefree of degree $2m + 1 \geq 5$, and let C be the smooth projective curve with function field

$$K(C) = K(x, y), \quad y^2 = f(x).$$

Let $\pi = x : C \longrightarrow \mathbb{P}^1$.

(a) Prove that $\deg \pi = 2$. Determine the ramification points of π and show that every ramification index is equal to 2.

(b) Apply the Riemann–Hurwitz formula to prove that $g(C) = m$.

(c) Let P_∞ be the unique point lying over $\infty \in \mathbb{P}^1$. Prove that

$$\operatorname{div}\left(\frac{dx}{y}\right) = (2m - 2)[P_\infty].$$

(d) Prove that

$$\frac{dx}{y}, \quad x\frac{dx}{y}, \quad \ldots, \quad x^{m-1}\frac{dx}{y}$$

form a basis for the space of regular differentials on C.